# Supplementary Material

*How Much Reported Breast Cancer AI Accuracy Survives Honest Validation? An Optimism-Ladder Audit of Metastasis Classifiers and an Audit-Ready Digital Twin*

Every number in this supplement is produced by the released code: *CaseStudy1_MetastasisLadder.ipynb* (public GEO data; random seed 20260806), *PECT_DigitalTwin_Audit_v3.ipynb* (digitised PECT data, released as CSV files; a separate random generator for each analysis) and *revision_analyses.py* (bootstrap intervals for the calibration slopes, comparison of included and excluded patients, and Fig. 1). Tables I–VII are in the main text; Tables S1–S14 are below. In the tables, "5-yr DM" is five-year distant metastasis.

## S1. Case study 1: patient flow (TRIPOD+AI)

**Table S1.** Patient flow and comparison of included and excluded patients. (a) Patient flow per cohort. GSE2603 contains 121 arrays, of which 22 cell-line samples were removed. Five-year distant metastasis was coded 1 for an event within 60 months and 0 for follow-up of at least 60 months without an event; patients censored earlier were excluded (GSE7390 and GSE11121: t.dmfs/e.dmfs; GSE2603: the authors' binary five-year field). Bone relapse at any time is recorded only in GSE2034 and GSE2603; in GSE2603, 17 of the 99 patients have no bone-relapse status.

| Cohort | Primary-tumour arrays | 5-yr DM evaluable | 5-yr DM events | Excluded (censored < 5 y) | Bone relapse evaluable | Bone relapse events | Size and grade complete |
|---|---|---|---|---|---|---|---|
| GSE2034 | 286 | 0 | 0 | – | 286 | 69 | 0 |
| GSE11121 | 200 | 182 | 28 | 18 | 0 | 0 | 200 |
| GSE2603 | 99 | 68 | 22 | 31 | 82 | 14 | 0 |
| GSE7390 | 198 | 190 | 36 | 8 | 0 | 0 | 196 |

**Table S1 (continued).** (b) Patients included in the five-year distant-metastasis analysis versus patients excluded because they were censored before 60 months, for every clinical variable recorded in the series matrices. In GSE7390, grade is missing for two included patients. All $p \geq 0.16$, although the excluded groups are small.

| Cohort | Variable | Included | Excluded | p | Test |
|---|---|---|---|---|---|
| GSE11121 (182 included, 18 excluded) | Tumour size, cm (median) | 1.9 | 2.0 | 0.160 | Mann–Whitney |
| | Grade 1/2/3 | 27/122/33 | 2/14/2 | 0.639 | $\chi^2$ |
| GSE7390 (190 included, 8 excluded) | Age, years (median) | 46.5 | 43.5 | 0.196 | Mann–Whitney |
| | Tumour size, cm (median) | 2.0 | 2.8 | 0.292 | Mann–Whitney |
| | Grade 1/2/3 | 27/81/80 | 3/2/3 | 0.190 | $\chi^2$ |
| | ER negative/positive | 61/129 | 3/5 | 0.715 | Fisher |
| GSE2603 (68 included, 31 excluded) | Age at diagnosis, years (median) | 56.0 | 56.0 | 0.757 | Mann–Whitney |
| | Tumour size, cm (median) | 3.0 | 3.5 | 0.566 | Mann–Whitney |
| | Positive lymph nodes (median) | 1.0 | 2.0 | 0.252 | Mann–Whitney |
| | ER negative/positive | 29/39 | 13/18 | 1.000 | Fisher |

## S2. Case study 1: preprocessing, models and protocols

**Genes.** GPL96 probes annotated to exactly one gene symbol were kept, and for each gene the probe with the highest mean log2 expression over all 783 arrays (computed without any outcome) represents the gene, giving 12,502 genes. The GSE2034, GSE11121 and GSE2603 series matrices contain MAS5 intensities (transformed as log2 of max(x, 1)); GSE7390 is supplied on the log2 scale.

**Fold-internal preprocessing.** Quantile normalisation to the mean sorted training distribution (each sample mapped individually); parametric empirical-Bayes ComBat fitted on the training cohorts; held-out samples from a cohort present in training are adjusted with that cohort's estimates, whereas an unseen cohort is not batch-adjusted; genes standardised with training means and SDs; 50 genes selected by |AUC − 0.5| on the training samples.

**Models.** All settings were fixed before any model was fitted, and nothing was tuned inside the training folds (scikit-learn 1.8; all random seeds derived from 20260806). Penalised logistic regression: L2 penalty, C = 0.1, balanced class weights, up to 2,000 iterations. SVM: RBF kernel, C = 1, $\gamma$ = 'scale', Platt probabilities, balanced class weights. Random forest: 400 trees, minimum leaf size 3, $\sqrt{p}$ features per split, balanced class weights. Gradient boosting (used instead of XGBoost): 500 trees, learning rate 0.03, depth 3, subsample 0.8, 60% of features per split, balanced sample weights. Histogram gradient boosting (used instead of LightGBM): 500 iterations, learning rate 0.03, 7 leaves, depth 3, L2 penalty 2, balanced class weights. Feed-forward network: hidden layers of 64 and 32 units, ReLU activation, Adam solver, weight decay ($\alpha$) 0.01, early stopping on a 10% validation split, up to 500 iterations, no class weighting. Clinical comparator: logistic regression (C = 1, balanced class weights) on standardised tumour size and grade.

**Protocols.** L1, leave-one-cohort-out (the train/test matrix is Table II of the main text); L2, four repeats of stratified five-fold cross-validation with preprocessing and gene selection inside the folds; L3, as L2 but with the 50 genes selected once on all samples; L4, 20 stratified random 80/20 splits with genes selected on all samples, keeping the best of the six families in each split; L5, resubstitution. Sensitivity analyses for L1: each unseen test cohort centred and scaled per gene on its own unlabelled data; and quantile normalisation plus ComBat fitted on all 783 arrays before splitting.

**AUC summaries.** The *per-cohort AUC* is measured on one held-out cohort, with a DeLong variance. The *random-effects AUC* pools the per-cohort AUCs of one family by DerSimonian–Laird weighting on the logit scale; $I^2$ and the between-cohort variance $\tau^2$ are on the same logit scale. The *mean AUC* used for the ladder is an unweighted mean, first over held-out cohorts (L1), over the 20 folds (L2, L3) or over the 20 splits (L4), and then over the six families; at L4 it averages the best family in each split, and at L5 each family contributes one in-sample AUC. The *cohort-averaged AUC* of a single model (a family under strict L1, the clinical model or a baseline) is the unweighted mean of its per-cohort AUCs; for the random gene sets it is also averaged over the 60 sets. In Fig. 1a the error bars are, for L1, the mean across families of each family's random-effects 95% CI limits; for L2–L4, the 2.5th–97.5th percentiles over folds or splits; and for L5, the range across families.

**Table S2.** AUC by model family and rung. L1 columns are cohort-averaged AUCs under the strict pipeline, with each test cohort standardised on its own data, and with global quantile normalisation plus ComBat; L2 and L3 are means over 20 folds; the L4 column gives each family's mean over 20 splits, and the row below it gives the ladder value (best family in each split); L5 is the in-sample AUC.

| Model | L1 strict | L1 test cohort self-standardised | L1 global harmonisation | L2 | L3 | L4 | L5 |
|---|---|---|---|---|---|---|---|
| ***5-yr distant metastasis*** | | | | | | | |
| Penalised logistic regression | 0.635 | 0.635 | 0.635 | 0.663 | 0.741 | 0.715 | 0.852 |
| SVM (RBF) | 0.619 | 0.619 | 0.617 | 0.662 | 0.743 | 0.741 | 0.955 |
| Random forest | 0.657 | 0.622 | 0.618 | 0.682 | 0.730 | 0.727 | 1.000 |
| Gradient boosting | 0.628 | 0.578 | 0.577 | 0.639 | 0.696 | 0.664 | 1.000 |
| Histogram gradient boosting | 0.631 | 0.595 | 0.598 | 0.638 | 0.683 | 0.664 | 1.000 |
| Neural network | 0.633 | 0.621 | 0.613 | 0.632 | 0.618 | 0.619 | 0.601 |
| **Mean AUC over six families** | 0.634 | 0.612 | 0.610 | 0.653 | 0.702 | – | 0.901 |
| **L4: best of six per split, mean of 20 splits** | | | | | | 0.757 | |
| ***Bone relapse*** | | | | | | | |
| Penalised logistic regression | 0.588 | 0.597 | 0.594 | 0.670 | 0.799 | 0.813 | 0.907 |
| SVM (RBF) | 0.568 | 0.568 | 0.573 | 0.657 | 0.799 | 0.811 | 0.987 |
| Random forest | 0.562 | 0.555 | 0.553 | 0.652 | 0.804 | 0.820 | 1.000 |
| Gradient boosting | 0.585 | 0.579 | 0.576 | 0.658 | 0.795 | 0.804 | 1.000 |
| Histogram gradient boosting | 0.565 | 0.567 | 0.544 | 0.647 | 0.790 | 0.782 | 1.000 |
| Neural network | 0.577 | 0.601 | 0.604 | 0.623 | 0.744 | 0.727 | 0.740 |
| **Mean AUC over six families** | 0.574 | 0.578 | 0.574 | 0.651 | 0.789 | – | 0.939 |
| **L4: best of six per split, mean of 20 splits** | | | | | | 0.834 | |

**Table S3.** Strict leave-one-cohort-out AUC per held-out cohort (DeLong 95% CI) and DerSimonian–Laird random-effects AUC. For bone relapse each estimate pools only two cohorts. $I^2$ and $\tau^2$ refer to the logit scale and are imprecise with two or three cohorts.

| Model | Held-out cohort 1 | Held-out cohort 2 | Held-out cohort 3 | Random-effects AUC (95% CI) | $I^2$ (%) | $\tau^2$ |
|---|---|---|---|---|---|---|
| ***5-yr distant metastasis (GSE11121: 182 patients, 28 events, GSE7390: 190 patients, 36 events, GSE2603: 68 patients, 22 events)*** | | | | | | |
| Penalised logistic regression | 0.671 (0.556–0.786) | 0.615 (0.517–0.713) | 0.619 (0.478–0.759) | 0.633 (0.565–0.696) | 0 | 0.000 |
| SVM (RBF) | 0.671 (0.559–0.784) | 0.635 (0.531–0.740) | 0.549 (0.406–0.693) | 0.626 (0.556–0.692) | 0 | 0.000 |
| Random forest | 0.696 (0.591–0.801) | 0.669 (0.580–0.758) | 0.605 (0.467–0.742) | 0.663 (0.600–0.722) | 0 | 0.000 |
| Gradient boosting | 0.660 (0.551–0.769) | 0.654 (0.558–0.750) | 0.570 (0.424–0.716) | 0.638 (0.571–0.700) | 0 | 0.000 |
| Histogram gradient boosting | 0.670 (0.571–0.770) | 0.621 (0.525–0.716) | 0.603 (0.464–0.742) | 0.635 (0.571–0.694) | 0 | 0.000 |
| Neural network | 0.680 (0.554–0.805) | 0.667 (0.588–0.746) | 0.552 (0.411–0.693) | 0.644 (0.573–0.709) | 15 | 0.011 |
| ***Bone relapse (GSE2034: 286 patients, 69 events, GSE2603: 82 patients, 14 events)*** | | | | | | |
| Penalised logistic regression | 0.565 (0.488–0.642) | 0.610 (0.473–0.748) | | 0.576 (0.507–0.641) | 0 | 0.000 |
| SVM (RBF) | 0.559 (0.480–0.638) | 0.578 (0.428–0.728) | | 0.563 (0.493–0.631) | 0 | 0.000 |
| Random forest | 0.558 (0.483–0.632) | 0.565 (0.427–0.703) | | 0.560 (0.493–0.624) | 0 | 0.000 |
| Gradient boosting | 0.566 (0.493–0.640) | 0.603 (0.454–0.752) | | 0.573 (0.506–0.638) | 0 | 0.000 |
| Histogram gradient boosting | 0.545 (0.473–0.617) | 0.585 (0.432–0.738) | | 0.552 (0.487–0.616) | 0 | 0.000 |
| Neural network | 0.548 (0.468–0.627) | 0.606 (0.472–0.740) | | 0.562 (0.493–0.629) | 0 | 0.000 |

Cohort order: 5-yr DM, GSE11121 / GSE7390 / GSE2603; bone relapse, GSE2034 / GSE2603.

**Table S4.** Paired DeLong comparisons with penalised logistic regression within each held-out cohort (strict L1), Holm-adjusted within endpoint and sorted by unadjusted p.

| Model | Held-out cohort | AUC difference (logistic regression − model) | p | Holm-adjusted p |
|---|---|---|---|---|
| ***5-yr distant metastasis*** | | | | |
| Random forest | GSE7390 | −0.054 | 0.096 | 1.000 |
| SVM (RBF) | GSE2603 | 0.069 | 0.124 | 1.000 |
| Neural network | GSE7390 | −0.052 | 0.214 | 1.000 |
| Gradient boosting | GSE7390 | −0.039 | 0.297 | 1.000 |
| SVM (RBF) | GSE7390 | −0.020 | 0.396 | 1.000 |
| Random forest | GSE11121 | −0.025 | 0.409 | 1.000 |
| Neural network | GSE2603 | 0.066 | 0.485 | 1.000 |
| Gradient boosting | GSE2603 | 0.048 | 0.488 | 1.000 |
| Gradient boosting | GSE11121 | 0.011 | 0.782 | 1.000 |
| Histogram gradient boosting | GSE2603 | 0.016 | 0.812 | 1.000 |
| Random forest | GSE2603 | 0.014 | 0.816 | 1.000 |
| Histogram gradient boosting | GSE7390 | −0.006 | 0.866 | 1.000 |
| Neural network | GSE11121 | −0.009 | 0.870 | 1.000 |

| Model | Held-out cohort | AUC difference (logistic regression − model) | p | Holm-adjusted p |
|---|---|---|---|---|
| SVM (RBF) | GSE11121 | −0.001 | 0.980 | 1.000 |
| Histogram gradient boosting | GSE11121 | 0.000 | 0.995 | 1.000 |
| ***Bone relapse*** | | | | |
| Random forest | GSE2603 | 0.045 | 0.242 | 1.000 |
| SVM (RBF) | GSE2603 | 0.033 | 0.467 | 1.000 |
| Histogram gradient boosting | GSE2034 | 0.020 | 0.474 | 1.000 |
| Histogram gradient boosting | GSE2603 | 0.025 | 0.625 | 1.000 |
| Random forest | GSE2034 | 0.007 | 0.629 | 1.000 |
| SVM (RBF) | GSE2034 | 0.006 | 0.662 | 1.000 |
| Neural network | GSE2034 | 0.018 | 0.793 | 1.000 |
| Gradient boosting | GSE2603 | 0.007 | 0.888 | 1.000 |
| Gradient boosting | GSE2034 | −0.001 | 0.951 | 1.000 |
| Neural network | GSE2603 | 0.004 | 0.962 | 1.000 |

**Table S5.** External calibration under strict L1 for each held-out cohort: calibration slope (ideal value 1) and calibration-in-the-large intercept (ideal value 0). For penalised logistic regression, the 95% CI of the slope comes from 2,000 bootstrap resamples of patients within the held-out cohort (Table IV of the main text). Class weighting shifts predicted probabilities upwards, which contributes to the negative intercepts of the weighted models.

| Model | Held-out cohort | Slope | Slope 95% CI | Intercept |
|---|---|---|---|---|
| ***5-yr distant metastasis*** | | | | |
| Penalised logistic regression | GSE11121 | 0.369 | 0.11 to 0.66 | −1.354 |
| Penalised logistic regression | GSE2603 | 0.321 | −0.03 to 0.77 | −1.478 |
| Penalised logistic regression | GSE7390 | 0.228 | 0.04 to 0.43 | −2.187 |
| SVM (RBF) | GSE11121 | 0.543 | | −0.213 |
| SVM (RBF) | GSE2603 | 0.311 | | 0.667 |
| SVM (RBF) | GSE7390 | 0.485 | | −0.381 |
| Random forest | GSE11121 | 0.613 | | −0.312 |
| Random forest | GSE2603 | 0.533 | | 0.412 |
| Random forest | GSE7390 | 0.637 | | −0.675 |
| Gradient boosting | GSE11121 | 0.175 | | 0.077 |
| Gradient boosting | GSE2603 | 0.123 | | 1.473 |
| Gradient boosting | GSE7390 | 0.204 | | −0.480 |
| Histogram gradient boosting | GSE11121 | 0.264 | | −0.126 |
| Histogram gradient boosting | GSE2603 | 0.155 | | 0.971 |
| Histogram gradient boosting | GSE7390 | 0.181 | | −0.396 |
| Neural network | GSE11121 | 0.691 | | −0.411 |
| Neural network | GSE2603 | 0.368 | | 0.098 |
| Neural network | GSE7390 | 0.767 | | −0.572 |
| ***Bone relapse*** | | | | |
| Penalised logistic regression | GSE2034 | 0.110 | −0.02 to 0.25 | −1.132 |

| Model | Held-out cohort | Slope | Slope 95% CI | Intercept |
|---|---|---|---|---|
| Penalised logistic regression | GSE2603 | 0.160 | −0.10 to 0.46 | −0.859 |
| SVM (RBF) | GSE2034 | 0.138 | | 0.216 |
| SVM (RBF) | GSE2603 | 0.084 | | 0.101 |
| Random forest | GSE2034 | 0.227 | | −0.306 |
| Random forest | GSE2603 | 0.135 | | −0.081 |
| Gradient boosting | GSE2034 | 0.039 | | 2.250 |
| Gradient boosting | GSE2603 | 0.130 | | 0.505 |
| Histogram gradient boosting | GSE2034 | 0.056 | | −0.424 |
| Histogram gradient boosting | GSE2603 | 0.137 | | −0.125 |
| Neural network | GSE2034 | 0.531 | | −0.813 |
| Neural network | GSE2603 | 0.436 | | −1.227 |

**Table S6.** Decision-curve analysis on the pooled strict-L1 predictions (threshold grid 0.05–0.60 in steps of 0.025): thresholds at which the net benefit of the model exceeds both treat-all and treat-none. These ranges apply to the event rate shown and would shift in a population with a different baseline risk.

| Model | Thresholds with net benefit above treat-all and treat-none |
|---|---|
| ***5-yr distant metastasis (event rate 19.5%; 440 patients, 86 events)*** | |
| Penalised logistic regression | 0.075–0.225 |
| SVM (RBF) | 0.100–0.300 |
| Random forest | 0.075–0.275 |
| Gradient boosting | 0.175–0.300 |
| Histogram gradient boosting | 0.175–0.250 |
| Neural network | 0.100–0.300 |
| ***Bone relapse (event rate 22.6%; 368 patients, 83 events)*** | |
| Penalised logistic regression | 0.175–0.250 |
| SVM (RBF) | 0.200–0.275 |
| Random forest | 0.050, 0.100, 0.175–0.250 |
| Gradient boosting | none |
| Histogram gradient boosting | 0.225–0.250 |
| Neural network | 0.175–0.225 |

**Table S7.** Baselines under strict L1 (cohort-averaged AUC) and signature stability over 50 stratified 80% subsamples drawn without replacement. Proliferation score: unfitted mean z-score of BIRC5, CCNB1, CDC20, CEP55, NDC80, MKI67, PTTG1, RRM2, TYMS and UBE2C (NUF2 is absent from HG-U133A). Random sets: 60 sets of 50 genes, each fitted with penalised logistic regression.

| Endpoint | Proliferation score | Random 50-gene sets | Learned penalised logistic regression | Jaccard index, mean (SD) | Distinct genes selected | Genes in ≥ 80% of lists | Genes in all lists |
|---|---|---|---|---|---|---|---|
| 5-yr distant metastasis | 0.654 | 0.577 | 0.635 | 0.38 (0.06) | 230 | 18 | 4 |
| Bone relapse | 0.475 | 0.524 | 0.588 | 0.22 (0.05) | 384 | 5 | 1 |

**Table S8.** Clinical model (tumour size and grade) versus transcriptomic models trained and tested on the same complete-case patients (five-year distant metastasis; GSE2603 lacks grade).

| Held-out cohort | Patients (events) | Size and grade | Penalised LR | SVM (RBF) | Random forest | Gradient boosting | Hist. gradient boosting | Neural network |
|---|---|---|---|---|---|---|---|---|
| GSE11121 | 182 (28) | 0.723 | 0.662 | 0.623 | 0.692 | 0.633 | 0.672 | 0.661 |
| GSE7390 | 188 (36) | 0.662 | 0.583 | 0.651 | 0.643 | 0.624 | 0.570 | 0.678 |

## S3. Case study 2: digital twin

**Measurement error.** Day 0 was digitised twice in the source figures; the duplicate readings give a digitisation SD of 0.0038 on the log scale, and their geometric mean is used as the single day-0 value. The pooled residual SD of the three pre-treatment scans is $\sigma = 0.1202$ (11 df). The noise variance of the observed kill rate, including the day-0 scan shared by the slope and the first treatment interval, is 0.000995.

**Forecast-error variance.** With OLS slope weights $w_t$ over $t = (-30, -15, 0)$, measurement errors $e_t \sim N(0, \sigma^2)$, prior $k_1 \sim N(\mu, \tau^2)$, late change $d = k_2 - k_1 \sim N(\delta, \sigma_d^2)$ and shrinkage weight $K$, the forecast error is $y_{12} - \hat{y}_{12} = e_{12} - (1 + K)e_6 + [K - 6(1 - K)w_0]e_0 - 6(1 - K)(w_{-30}e_{-30} + w_{-15}e_{-15}) - 6(1 - K)(k_1 - \mu) - 6(d - \delta)$, so that $\mathrm{Var} = \sigma^2[1 + (1 + K)^2 + (K - 6(1 - K)w_0)^2 + 36(1 - K)^2(w_{-30}^2 + w_{-15}^2)] + 36(1 - K)^2(\tau^2 + \mathrm{Var}\,\hat{\mu}) + 36(\sigma_d^2 + \mathrm{Var}\,\delta)$. A Monte Carlo check with 200,000 replicates agreed with this expression to within 0.4% for $K = 0$, 0.5 and 1.

**Nested ladder and uncertainty.** L3 tunes the prior scale over six values (0.25, 0.5, 1, 2, 4, 8) and the drift weight over five values (0, 0.5, 1, 1.5, 2) on the evaluation lesions; L4 then chooses one model variant for all lesions by its evaluation error. Lesion-bootstrap 95% CIs use 4,000 resamples of the 11 lesions (percentile method) and treat lesions as independent. With two patients a patient-clustered bootstrap has only three distinct resamples, so the L1 error of the twin is also reported for each patient: MdAPE 17.3% for patient 2 (5 lesions) and 34.6% for patient 3 (6 lesions).

**Table S9.** All models under leave-one-patient-out (L1) and leave-one-lesion-out (L2). Rate-based models use the analytic predictive variance; LOCF and growth only use a Gaussian whose SD is the RMS log error on the training lesions. Interval score and CRPS are on the log scale.

| Model | MdAPE (%) | RMS log error | RMSE ($mm^3$) | 95% coverage (%) | Interval width (hi/lo ratio) | Interval score | CRPS |
|---|---|---|---|---|---|---|---|
| ***Leave-one-patient-out (L1)*** | | | | | | | |
| Last observation carried forward (LOCF) | 61.9 | 0.687 | 232.9 | 90.9 | 20.7 | 3.73 | 0.393 |
| Growth only (no treatment effect) | 75.9 | 0.728 | 265.1 | 90.9 | 27.2 | 4.13 | 0.422 |
| Lesion-only rate (static k) | 28.6 | 0.494 | 174.6 | 90.9 | 8.3 | 2.56 | 0.280 |
| Lesion-only rate + drift | 27.1 | 0.449 | 132.2 | 100.0 | 10.2 | 2.10 | 0.263 |
| Population rate, no drift | 38.5 | 0.539 | 146.0 | 90.9 | 8.9 | 3.84 | 0.304 |
| Population rate + drift | 21.3 | 0.527 | 108.0 | 90.9 | 10.8 | 3.10 | 0.293 |
| Twin without drift (Kalman only) | 32.4 | 0.509 | 136.7 | 90.9 | 7.8 | 3.55 | 0.281 |
| Digital twin (Kalman + drift) | 21.7 | 0.466 | 96.9 | 90.9 | 9.6 | 2.82 | 0.261 |
| ***Leave-one-lesion-out (L2)*** | | | | | | | |
| Last observation carried forward (LOCF) | 61.9 | 0.687 | 232.9 | 90.9 | 15.8 | 4.06 | 0.400 |
| Growth only (no treatment effect) | 75.9 | 0.728 | 265.1 | 90.9 | 18.4 | 4.29 | 0.425 |
| Lesion-only rate (static k) | 28.6 | 0.494 | 174.6 | 81.8 | 6.4 | 3.17 | 0.282 |
| Lesion-only rate + drift | 33.1 | 0.484 | 134.8 | 90.9 | 7.0 | 2.08 | 0.286 |
| Population rate, no drift | 26.9 | 0.519 | 118.0 | 90.9 | 5.9 | 3.45 | 0.294 |
| Population rate + drift | 39.8 | 0.504 | 93.7 | 90.9 | 6.5 | 2.87 | 0.291 |
| Twin without drift (Kalman only) | 30.8 | 0.487 | 125.4 | 81.8 | 5.7 | 3.45 | 0.275 |
| Digital twin (Kalman + drift) | 31.4 | 0.471 | 97.0 | 90.9 | 6.3 | 2.57 | 0.275 |

**Table S10.** Nested twin ladder: MdAPE with lesion-bootstrap 95% CI. The L3 tuning chose prior scale 1 and drift weight 0.5; the L4 choice was the full twin. Prior variance per L1 fold: 0.00028, 0.00154 (the floor was never reached). Kalman gain range: L1 0.22–0.61, L2 0.36–0.53. L1 coverage: 10/11 (Clopper–Pearson 95% CI 58.7–99.8%).

| Rung | MdAPE (%) | Bootstrap 95% CI (%) | RMS log error |
|---|---|---|---|
| L1, leave one patient out | 21.7 | 15.0–49.0 | 0.466 |
| L2, leave one lesion out | 31.4 | 11.6–47.5 | 0.471 |
| L3, tuned on evaluation lesions | 25.2 | 18.6–44.5 | 0.465 |
| L4, best variant on evaluation lesions | 25.2 | 18.6–44.5 | 0.465 |
| L5, in-sample | 5.8 | 2.5–9.7 | 0.075 |

**Table S11.** Within-patient permutation test. Each lesion receives the day-6 response of another lesion from the same patient; all derangements within each patient are evaluated (44 for patient 2 × 265 for patient 3 = 11,660). One-sided p = (1 + number of rearrangements with MdAPE ≤ observed) / (1 + 11,660).

| Protocol | Twin MdAPE (‰) | Median MdAPE under rearrangement (‰) | p | Rearrangements |
|---|---|---|---|---|
| L1 | 21.7 | 34.7 | 0.083 | 11,660 |
| L2 | 31.4 | 35.1 | 0.150 | 11,660 |

**Table S12.** Paired Wilcoxon tests on absolute log errors. These are descriptive only, because lesions within a patient are not independent.

| Protocol | Comparison | p |
|---|---|---|
| L1 | Twin vs lesion-only rate | 1.000 |
| L1 | Twin vs lesion-only rate + drift | 0.765 |
| L1 | Twin vs population rate + drift | 0.320 |
| L1 | Twin vs twin without drift | 0.953 |
| L1 | Twin vs LOCF | 0.240 |
| L2 | Twin vs lesion-only rate | 0.765 |
| L2 | Twin vs lesion-only rate + drift | 0.700 |
| L2 | Twin vs population rate + drift | 1.000 |
| L2 | Twin vs twin without drift | 0.638 |
| L2 | Twin vs LOCF | 0.147 |

**Table S13.** Simulation of the test harness (300 synthetic cohorts per row; leave-one-patient-out folds; test statistic = mean absolute log error; 199 within-patient permutations per cohort). Scenarios: null, late kill rate independent of the early rate; as estimated from the 11 lesions; strong, residual change in rate halved; strong, precise, as strong with a measurement SD of 0.04. Entries are the percentage of cohorts with $p < 0.05$: false-positive rate in the null scenario and power in the others. The Wilcoxon test is one-sided, twin versus population rate.

| Patients | Lesions | Scenario | Permutation test (% rejected) | Wilcoxon test (% rejected) |
|---|---|---|---|---|
| 2 | 12 | Null | 3.0 | 4.7 |
| 2 | 12 | As estimated | 7.0 | 13.0 |
| 2 | 12 | Strong | 12.0 | 16.7 |
| 2 | 12 | Strong, precise | 51.7 | 38.3 |
| 4 | 20 | Null | 2.3 | 0.3 |
| 4 | 20 | As estimated | 7.3 | 8.7 |
| 4 | 20 | Strong | 14.0 | 13.3 |
| 4 | 20 | Strong, precise | 63.3 | 42.0 |
| 10 | 50 | Null | 3.3 | 0.3 |
| 10 | 50 | As estimated | 14.7 | 12.7 |
| 10 | 50 | Strong | 24.7 | 17.7 |
| 10 | 50 | Strong, precise | 97.7 | 75.0 |
| 20 | 100 | Null | 1.0 | 0.0 |
| 20 | 100 | As estimated | 24.3 | 12.3 |
| 20 | 100 | Strong | 40.0 | 22.3 |
| 20 | 100 | Strong, precise | 100.0 | 95.7 |
| 40 | 200 | Null | 0.3 | 0.0 |
| 40 | 200 | As estimated | 37.0 | 19.0 |
| 40 | 200 | Strong | 64.7 | 26.0 |
| 40 | 200 | Strong, precise | 100.0 | 99.7 |

**Table S14.** Illustrative cross-species prior (not a finding). Mouse fit: $k_{max} = 0.493$ per day; EC50 = 0.225 kV, below the lowest tested dose and therefore not identifiable; k(30 kV) = 0.490 per day, 13.0 times the median human early rate. Forecasts include the population drift and use L1 folds.

| Prior SD | Kalman gain | MdAPE (%) |
|---|---|---|
| 0.02 | 0.287 | 84.6 |
| 0.05 | 0.715 | 60.6 |
| 0.10 | 0.910 | 41.3 |
| 0.20 | 0.976 | 30.9 |